

of the score. The good thing about this method is that it is dynamic; it can be changed depending on the category of movies, the country or even with the demographic of users.

Striving for continuous improvement

These aren't the only forms of data that Netflix uses though. Through a continually evolving process the data scientists at Netflix keep varying the criteria for ranking and experiment with different variations. Their methodology of testing revolves around the same principles: Offline testing of



Some of the multi-audio, multi-caption language options on Netflix streaming

a hypothesis using archived data, selective testing of the data on online subscribers over months and, if the hypothesis proves to be a more successful scheme than the existing methodology, it is rolled out to all subscribers. Multiple metrics are tracked to measure the performance of any particular model, including metrics specifically

designed for this purpose such as the Root Mean Square Error (RMSE) which was popularised in the Netflix prize and is a measure of the variation of scores with diversity. The company also innovates with in-house competitions such as the Top10 Marathon, which was a focused 10 week period in which different algorithms were tested quickly, many of which are now part of the Netflix system.

As the number of countries in which Netflix operates grows, the demands of the data mining and machine learning algorithm increase exponentially. There is active research currently going on to develop algorithms that are diverse enough to integrate cultural awareness, enabling Netflix to truly establish itself as an international organisation.

Language is also a huge issue as the expansion of Netflix services into 21 different languages brings with it a host of challenges. At the same time, the availability of a service like Netflix ultimately leads to an expansion of tastes and an overall improvement in the internationalism of entertainment. The time is not too far away when one can expect to see Salman Khan